

LESSON 7-6 • SESSION 1 • GUIDED NOTES**Distance Between Points on a Grid**

Learning target – I can find the distance between two points that share a coordinate. · 6.NS.C.8

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
distance <i>distancia</i>	How far apart two points are. It is never ____.	(2, 1) to (2, 5) is 4 units
absolute value <i>valor absoluto</i>	How far a number is from ____, ignoring its sign.	$ -7 = 7$
horizontal distance <i>distancia horizontal</i>	Distance side to side. Use the ____-coordinates.	(-3, 2) to (4, 2)
vertical distance <i>distancia vertical</i>	Distance up and down. Use the ____-coordinates.	(5, -1) to (5, 6)

KEY IDEA*complete each sentence*

- If two points share a y-coordinate, find the distance with the ____-coordinates.
- If the changing coordinates have opposite signs, ____ their absolute values. If they have the same sign, ____ them.

WORKED EXAMPLE*fill in the blanks as we solve*

Find the distance from (-4, 3) to (5, 3).

Step 1 The points share the y-coordinate ____, so use the x-coordinates.**Step 2** -4 and 5 have _____ signs, so the trip crosses the y-axis.**Step 3** Add the absolute values: $4 + 5 =$ ____.**Answer** The distance is ____ units.**YOU TRY***use the example as a model***1.** Find the distance between each pair of points.**a.** (1, 6) and (1, -3) ____**b.** (-7, -2) and (-2, -2) ____**2.** Point A is at (3, -1). Point B is 5 units directly above A. Name point B.*Show your work.***B =** _____**SUM IT UP***in your own words*

To find the distance between two points that share a coordinate, I ____ because ____.

LESSON 7-6 • SESSION 2 • GUIDED NOTES

Comparing Distances on the Plane

Learning target – I can decide whether to add or subtract to find a distance, then compare two distances. · 6.NS.C.8

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
quadrant <i>cuadrante</i>	One of the four _____ of the coordinate plane.	$(-3, -5)$ is in Quadrant III
compare <i>comparar</i>	Decide which amount is _____ or less.	$12 \text{ units} > 9 \text{ units}$
scale <i>escala</i>	How much real length one grid _____ stands for.	$1 \text{ unit} = 5 \text{ meters}$

KEY IDEA*complete each sentence*

- Find _____ distance first. Then compare the two numbers.
- With a scale, change _____ distances to real units before you compare.

WORKED EXAMPLE*fill in the blanks as we solve*

$A = (1, 2)$, $B = (1, 9)$, $C = (-8, 2)$. Each unit is 5 meters. Is B or C farther from A?

Step 1 A to B: same x, so $9 - 2 = \underline{\hspace{1cm}}$ units = 35 m.

Step 2 A to C: same y, opposite signs, so $1 + 8 = \underline{\hspace{1cm}}$ units = _____ m.

Answer _____ is farther, by 10 m.

YOU TRY*use the example as a model*

1. Add or subtract? Find each distance.

a. $(-2, 3)$ to $(7, 3)$ _____

b. $(5, 3)$ to $(12, 3)$ _____

2. From $(4, -3)$, Ana walks to $(4, 6)$ and Ben walks to $(-6, -3)$. Who walks farther?

Show your work.

Farther _____

SUM IT UP*in your own words*

To compare two distances on the plane, I _____ because _____.
